Dhanush Amsanathan

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PROFILE

AI/ML Engineer with experience building production-grade Generative AI, Large Language Model (LLM), and multimodal AI applications. Experienced in developing real-time voice AI, AI-powered evaluation systems, automated report generation, conversational analytics, and sales training platforms. Skilled in Python, LLM integration, prompt engineering, AI model evaluation, speech-to-text, audio/video processing, API development, AI inference optimization, and cost optimization. Experienced in deploying and improving end-to-end AI pipelines for production applications.

SKILLS

PROGRAMMING: Python, SQL

GENERATIVE AI & LLM: Generative AI, LLMs, VLMs, Prompt Engineering, LLM Evaluation, RAG, AI Agents, Model Selection, Fine-Tuning, LoRA, Transformers

AI & MACHINE LEARNING: Machine Learning, Deep Learning, NLP, Multimodal AI, Computer Vision, TensorFlow, Scikit-Learn

SPEECH & REAL-TIME AI: Speech-to-Text (STT), Audio Processing, Voice AI, Real-Time Audio/Video Processing, Speaker Diarization

BACKEND & DATA: API Development, REST APIs, MySQL, Data Pipelines, ETL

AI OPTIMIZATION: Prompt Optimization, Token Optimization, AI Cost Optimization, Model Routing, Inference Optimization

TOOLS & LIBRARIES: Pandas, NumPy, OpenCV, MediaPipe, FFmpeg, Git, Docker

EXPERIENCE

Junior ML Engineer | Project Lead & Product Manager — AI Mentor

LMES Academy, Chennai | May 2025 – Present

- Built end-to-end ML pipelines for dataset preparation, preprocessing, Speech-to-Text (STT) model fine-tuning, evaluation, and deployment.
- Developed a VLM-based video analysis system to process video content and generate structured insights and reports.
- Built and enhanced an AI Mentor sales training platform with real-time voice roleplay, peer-to-peer practice, Training Phases, LMS assessments, AI evaluation, stage-wise scoring, and automated reports.
- Developed AI Analysis workflows for BANT, QA, CSAT, Refund Classification, Call Tagging, Pattern Analysis, Lead Qualification, Consolidated Reports, and AI Pitch Automation.
- Engineered LLM evaluation and report-generation pipelines using Gemini and OpenRouter, including prompt engineering, transcript/audio processing, speaker diarization, validation, and retry/recovery mechanisms.
- Optimized AI systems for cost, latency, and reliability through prompt caching, model routing, token optimization, provider fallback, and cost-efficient model selection.
- Developed real-time audio/video AI features including speech processing, camera/face monitoring, script-copy detection, audio recovery, fullscreen monitoring, and noise cancellation.
- Built AI-powered learning features including video transcription, flashcard generation, learner journey, coaching/retry moments, and practice roleplay.

Data Science Intern

ANB Systems, LLC (a DNV company) | Product Company
Chennai (On-site) | Nov 2024 – Feb 2025

- Developed data analysis and preprocessing pipelines using Python, Pandas, and NumPy for product analytics and reporting
- Worked on P&ID data annotation and object detection, preparing and labeling data for machine learning workflows.
- Built a real-time analytics dashboard integrated with eTRACK+, enabling automated data retrieval, visualization, and monitoring of key metrics.
- Worked with MongoDB, JSON, and REST APIs to retrieve, process, and analyze application data.
- Collaborated with teams to improve reporting workflows and data-driven decision-making.

FREELANCE EXPERIENCE

Freelance Developer — Full-Stack, ML & AI Projects

Delivered 40+ freelance projects across full-stack web development, machine learning, and AI applications.

Key Work:

- Developed **3 full ML/AI projects**, including classification models, prediction systems, and computer vision tasks.
- Built and deployed full-stack applications with frontend, backend, APIs, databases, and third-party integrations.
- Developed ML and AI solutions including classification, prediction, computer vision, NLP, and LLM-powered applications.
- Managed projects end-to-end, including requirements gathering, development, testing, deployment, and client communication.

PROJECTS

Real-Time Sign Language Recognition System — Final Year Project

- Developed a real-time sign language recognition system using MediaPipe hand landmarks and Machine Learning for gesture classification.
- Implemented real-time hand tracking and gesture recognition using 21 hand landmarks.
- Technologies: Python, MediaPipe, OpenCV, NumPy, Pandas, Scikit-Learn

Online Shopping Process Analytics — Final Year Project

- Analyzed customer purchase behavior, seasonality, and business patterns to derive actionable business insights.
- Used SQL to analyze sales data and support inventory planning and customer targeting.
- Technologies: MySQL, SQL, Data Analysis, Business Intelligence

EDUCATION

B.Tech. Artificial Intelligence and Data Science

KCG College of Technology, Chennai | 2021 – 2025

CERTIFICATIONS

- ML Practitioner Certificate — Dataiku
- Developer Certificate — Dataiku
- Python for Machine Learning — Retech Solutions
- Programming Fundamentals using Python — Infosys
- MySQL Fundamentals — Simplilearn

Award /POSITION OF RESPONSIBILITY

- Young Talent of the Year Award — Recognized at LMES Academy for outstanding performance and contribution to AI/ML projects.
- AI Mentor — Project Lead & Product Manager — Led product development, AI feature delivery, technical execution, and production improvements.
- Team Leader — Sign Language Recognition Project — Led a 3-member team during the final-year project.